

Distribution-Free Causal Inference for Semicontinuous Outcomes: A Two-Part Bayesian Causal Forest with Duan's Smearing

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Abstract. Estimating conditional average treatment effects (CATE) and average treatment effects (ATE) is challenging when the outcome is semicontinuous, exhibiting both a discrete point mass at zero and a skewed continuous distribution for positive values. Existing methods either ignore the zero mass, impose fragile parametric assumptions, or suffer from extreme computational costs and data augmentation noise. This paper proposes Zero-Inflated Continuous Bayesian Causal Forests with Duan's Smearing (ZIC-BCF-Smear), a decoupled two-part nonparametric estimator that fits a Probit BCF on the full sample to model participation and a Gaussian BCF strictly on the active subpopulation to model positive intensity. Subpopulation Propensity Adjustment (SPA) controls the selection (collider) bias induced by conditioning on active participation, and Duan's nonparametric smearing estimator re-transforms predictions to the response scale, bypassing the fragile scale multipliers of parametric models. A comprehensive simulation benchmark shows that ZIC-BCF-Smear matches near-correct parametric models without their assumptions and remains stable and calibrated where competitors either diverge or pathologically over-cover. The method is then applied to individual-participant data from the SPECTRUM randomized phase III head-and-neck cancer trial, estimating the causal effect of assignment to panitumumab plus chemotherapy on the number of days patients spend with severe toxicity, an outcome for which a quarter of patients contribute an exact zero. Panitumumab assignment raises that burden by about ten days per patient under the trial's own differential treatment duration, and the decoupled structure localizes the effect mainly in the positive-intensity part rather than in whether the reconstructed severe-AE burden is positive.

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1 Introduction

Semicontinuous data, characterized by a discrete point mass at exactly zero alongside a continuous, highly right-skewed positive distribution for non-zero values, are ubiquitous in scientific research. Such outcomes frequently arise in clinical trials (Albert and

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Shen, 2005), health economics (Deb and Norton, 2018), and environmental epidemiology (Gottard et al., 2012). Classic examples include medical expenditures, where a large portion of a study cohort incurs zero costs during a follow-up period while a minority incurs highly skewed positive costs (Duan et al., 1983; Manning and Mullahy, 2001); biomarker levels, such as cardiac troponin measurements that are zero for healthy individuals but positively skewed for patients suffering myocardial injury under radiation exposure (Kim et al., 2024); and substance abuse frequencies or environmental exposure levels.

In these settings, estimating heterogeneous treatment effects is critical for precision medicine and targeted policy interventions. Specifically, analysts seek to estimate the Conditional Average Treatment Effect (CATE), which represents the expected treatment effect conditional on a high-dimensional vector of pre-treatment baseline covariates. Identifying which subpopulations benefit from or are harmed by a treatment enables clinicians and policymakers to make optimal, individualized decisions.

However, estimating CATEs when the outcome is semicontinuous poses severe methodological challenges, and existing causal inference tools exhibit major limitations. First, single-part causal estimators, such as standard Bayesian Additive Regression Trees (BART) (Hill, 2011) or Bayesian Causal Forests (BCF) (Hahn et al., 2020), ignore the discrete point mass at zero. By forcing a single smooth regression function across the zero discontinuity, they suffer from substantial bias and collapse in credible interval coverage, especially under compound Poisson-Gamma (Tweedie) outcomes. Second, traditional parametric two-part hurdle models, which model participation and positive intensity separately using generalized linear models (GLMs) (Oganisian et al., 2021), require the analyst to specify a rigid parametric distribution (e.g., log-normal or Gamma) for the positive part. If this distribution is misspecified, the parametric scale re-transformation multiplier $\exp(\sigma^2/2)$ introduces severe systematic scale bias, inflating CATE and ATE estimates and inducing numerical instability or incorrect treatment effect ordering. Third, flexible Bayesian nonparametric (BNP) mixtures, such as Dirichlet-process mixtures of regressions (e.g., DPglm or enriched Dirichlet processes) (Hannah et al., 2011; Oganisian et al., 2021; Kim et al., 2024), are computationally prohibitive due to $O(N^3)$ complexity. More importantly, they rely on data augmentation to impute continuous latent outcomes for the zero observations at every MCMC iteration, which propagates simulated noise into the posterior, causing pathologically over-wide intervals (yielding degenerate 100% coverage) that fail to guide decision-making.

To resolve these limitations, the Zero-Inflated Continuous Bayesian Causal Forest with Duan’s Smearing (ZIC-BCF-Smear) is proposed. The proposed method is a two-part Bayesian forest framework that decouples the participation hurdle from the positive intensity process, fitting a Probit BCF on the full sample and a Gaussian BCF strictly on the active subpopulation. To control for the selection bias (collider bias) induced by conditioning on a post-treatment active status, Subpopulation Propensity Adjustment (SPA) is introduced, which uses the active-subset propensity score as a regularizing control covariate in the prognostic forest. Finally, the log-scale predictions are mapped back to the original response scale using Duan’s Non-Parametric Smearing Estimator (Duan, 1983). By calculating the smearing factor from empirical residuals, the scale transformation is calibrated without committing to a parametric outcome family, achieving the

flexibility of nonparametric methods and the stability of correctly specified parametric hurdle models.

1.1 Related Work

The proposed method lies at the intersection of three literature streams: Bayesian forest-based causal inference, semicontinuous/hurdle modeling, and Bayesian nonparametric causal estimation.

BART and BCF for Heterogeneous Effects: The use of ensemble tree methods for causal inference was pioneered by [Hill \(2011\)](#) using Bayesian Additive Regression Trees (BART) ([Chipman et al., 2010](#)). [Hahn et al. \(2020\)](#) introduced Bayesian Causal Forests (BCF), which isolates the prognostic effect from the treatment effect and incorporates the propensity score directly into the prognostic forest to control for regularization bias. While BCF is a state-of-the-art CATE learner, it is designed for continuous Gaussian outcomes and suffers under semicontinuous data. Other CATE learners, such as frequentist causal forests ([Wager and Athey, 2018](#)), generalized random forests ([Athey et al., 2019](#)), and metalearners ([Künzel et al., 2019](#)), similarly lack native mechanisms to accommodate the zero spike without introducing bias or scaling instabilities.

Semicontinuous and Hurdle Modeling: Two-part hurdle models have a long history in statistics and econometrics, beginning with Tobit models ([Tobin, 1958](#)) and Cragg’s double-hurdle specification. [Duan et al. \(1983\)](#) and [Mullahy \(1986\)](#) formalized two-part models for medical cost estimation and count data, while [Olsen and Schafer \(2001\)](#) extended them to random-effects settings. In causal inference, zero-inflated modeling has been applied to evaluate average effects ([Ghosh et al., 2006](#)), zero-inflated mediation paths ([Cheng et al., 2018](#)), and subpopulation effects ([Wang and Albert, 2012](#)). However, these approaches rely on parametric GLMs and linear treatment effects, which fail to capture high-dimensional non-linear treatment heterogeneity.

Bayesian Nonparametrics in Causal Inference: To avoid parametric assumptions, several authors have developed Bayesian nonparametric (BNP) causal estimators. [Roy et al. \(2017\)](#) and [Xu et al. \(2018\)](#) developed BNP models for marginal structural models and quantile treatment effects. For semicontinuous outcomes, [Oganisian et al. \(2021\)](#) proposed DPglm, a Dirichlet-process mixture of generalized linear models, while [Kim et al. \(2024\)](#) introduced EDP-ZI, an enriched Dirichlet process mixture designed to handle covariate overlap issues. Although theoretically flexible, these mixtures require expensive MCMC updates and suffer from variance inflation due to continuous outcome imputation for the zeros, limiting their utility in practice.

1.2 Contributions

This work makes the following four key contributions:

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1. **Decoupled Hurdle-Intensity BCF Framework:** A two-part Bayesian Causal Forest model is developed that separates the zero-inflation hurdle (Probit BCF) from the continuous positive intensity (Gaussian BCF fit strictly on active units), avoiding data augmentation noise and posterior variance inflation.
2. **Subpopulation Propensity Adjustment (SPA):** A propensity-score adjustment technique strictly for the active subpopulation is proposed and formulated. This controls for the collider/selection bias that naturally arises when conditioning on post-treatment active status.
3. **Non-Parametric Duan’s Smearing Re-transformation:** Duan’s smearing estimator is incorporated within the BCF posterior draws to map predictions back to the original response scale. This replaces the fragile parametric scale multiplier $\exp(\sigma^2/2)$ with a distribution-free multiplier, resolving scale misspecification bias under non-log-normal intensities.
4. **New Clinical Knowledge on Panitumumab-Associated Toxicity Burden:** The first causal analysis of the SPECTRUM randomized phase III trial ([Vermorken et al., 2013](#)) on a severe-toxicity *duration* endpoint is delivered, estimating both the average and the conditional average effect of adding panitumumab to cisplatin and fluorouracil on the number of days a patient spends with grade 3 or higher toxicity. Published SPECTRUM safety reporting summarizes event incidence, grade, and category; no prior analysis estimates the causal effect of randomized panitumumab assignment on how many days of severe toxicity a patient actually accrues, nor how that burden varies with baseline characteristics. Adding panitumumab is found to raise the burden by roughly ten overlap-adjusted severe-toxicity days per patient, a conclusion that survives the change of outcome definition, and the two-part decomposition localizes the effect mainly in the positive-intensity part rather than in whether the reconstructed severe-AE burden is positive, which the aggregate incidence tables of the parent trial cannot separate.

Supporting these contributions, the estimator is validated in the most extensive simulation benchmark to date for semicontinuous heterogeneous treatment effects, comparing 5 estimators across 3 DGP families (Log-normal, Gamma, Tweedie), sample sizes $N \in \{100, 250, 500, 1000\}$, zero-inflation levels from 3% to 85%, and 100 independent seeds under both link-linear and link-nonlinear scenarios; and a second applied study on dental expenditures in the Medical Expenditure Panel Survey ([Agency for Healthcare Research and Quality, 2023](#)) serves as an external check against a mature applied literature, showing that ZIC-BCF-Smear recovers documented coverage-effect gradients on the expenditure scale that a parametric Gamma hurdle benchmark does not.

1.3 Paper Organization

The remainder of the paper is organized as follows. Section 2 formalizes semicontinuous potential outcomes, causal targets, and identification assumptions. Section 3 introduces the proposed ZIC-BCF-Smear model, subpopulation propensity adjustment,

and Duan’s smearing re-transformation algorithm. Section 4 presents the simulation framework, DGPs, and comparative results under link-linear and link-nonlinear data-generating processes. Section 5 applies the estimator to severe-adverse-event burden in a randomized head-and-neck cancer trial and summarizes a second application to dental expenditures. Section 6 states the limitations of the framework, and Section 7 concludes. Extended derivations, the complete simulation results, and the full versions of both applied studies are provided in the Supplementary Material (Souto et al., 2026).

2 Semicontinuous Causal Inference

Semicontinuous data are characterized by a discrete point mass at exactly zero alongside a continuous, typically highly right-skewed, positive distribution for non-zero values. Such outcomes are common in clinical trials, economic evaluations, and observational studies (e.g., medical expenditures (Deb and Norton, 2018), substance abuse frequency (Burgette et al., 2021), biomarker levels (Rustand et al., 2023), and environmental exposures (Friesen et al., 2021)). In this section, the semicontinuous causal inference framework is formalized under the potential outcomes setup.

2.1 Outcome Decomposition

Let $Y_i \geq 0$ denote the observed semicontinuous outcome for unit $i \in \{1, \dots, N\}$. Any semicontinuous outcome can be structurally decomposed into two parts:

$$Y_i = I_i \cdot Y_i^+ \quad (2.1)$$

where:

- $I_i = \mathbb{I}(Y_i > 0) \in \{0, 1\}$ is a binary participation or response indicator ($I_i = 1$ if the hurdle is cleared, and $I_i = 0$ if the outcome is zero).
- $Y_i^+ > 0$ represents the continuous response intensity, which is only defined or observed when the hurdle is cleared ($I_i = 1$).

2.2 Semicontinuous Potential Outcomes

Under the Rubin Causal Model framework (Rubin, 1974, 1978, 2005), let $Z_i \in \{0, 1\}$ represent the binary treatment assignment. For each unit i , the joint potential outcomes are defined as:

- Hurdle potential outcomes: $I_i(0), I_i(1) \in \{0, 1\}$, representing participation status under control and treatment, respectively.
- Intensity potential outcomes: $Y_i^+(0), Y_i^+(1) > 0$, representing positive intensity under control and treatment, respectively.

The potential outcomes on the raw semicontinuous scale are then defined as:

$$Y_i(z) = I_i(z) \cdot Y_i^+(z) \quad \text{for } z \in \{0, 1\} \quad (2.2)$$

2.3 Causal Estimands

In clinical and economic settings, the primary target of causal inference is the Conditional Average Treatment Effect (CATE) on the raw response scale:

$$\tau_{\text{resp}}(X_i) = E[Y_i(1) - Y_i(0) \mid X_i] \quad (2.3)$$

where X_i is a vector of pre-treatment baseline covariates.

Using the law of total expectation, the conditional expectation of the potential outcomes can be expressed as:

$$E[Y_i(z) \mid X_i] = P(I_i(z) = 1 \mid X_i) \cdot E[Y_i^+(z) \mid X_i, I_i(z) = 1] \quad (2.4)$$

Thus, the CATE on the raw response scale decomposes into the difference of two products, representing the participation probability and the conditional intensity:

$$\tau_{\text{resp}}(X_i) = P(I_i(1) = 1 \mid X_i)E[Y_i^+(1) \mid X_i, I_i(1) = 1] - P(I_i(0) = 1 \mid X_i)E[Y_i^+(0) \mid X_i, I_i(0) = 1] \quad (2.5)$$

The population Average Treatment Effect (ATE) is defined by integrating $\tau_{\text{resp}}(X_i)$ over the marginal covariate distribution of X :

$$\text{ATE} = E[\tau_{\text{resp}}(X_i)] = E[Y_i(1) - Y_i(0)] \quad (2.6)$$

2.4 Identification Assumptions

To identify the causal estimands from observational data, several standard assumptions are required (Rosenbaum and Rubin, 1983; Imbens and Rubin, 2015):

Assumption 2.1 (Stable Unit Treatment Value Assumption (SUTVA)). The potential outcomes for any unit are independent of the treatment assignments of other units (no interference), and there are no hidden versions of the treatment.

Assumption 2.2 (Strong Ignorability). Conditional on baseline covariates X_i , the treatment assignment Z_i is independent of the potential outcomes:

$$(Y_i(0), Y_i(1), I_i(0), I_i(1)) \perp\!\!\!\perp Z_i \mid X_i \quad (2.7)$$

Assumption 2.3 (Consistency). The observed outcome matches the potential outcome under the assigned treatment:

$$Y_i = Z_i Y_i(1) + (1 - Z_i) Y_i(0) \quad (2.8)$$

and similarly, $I_i = Z_i I_i(1) + (1 - Z_i) I_i(0)$ (VanderWeele, 2009).

Assumption 2.4 (Positivity / Overlap). The propensity score $\pi(X_i) = P(Z_i = 1 \mid X_i)$ is strictly bounded away from the boundaries:

$$\delta < \pi(X_i) < 1 - \delta \quad \text{for some } \delta > 0 \quad (2.9)$$

This ensures that every unit has a non-zero probability of receiving either treatment.

The overlap assumption is crucial for nonparametric estimators. While some Bayesian nonparametric methods (Kim et al., 2024) present their flexibility as a way to handle poor overlap, modeling the response surfaces directly with Bayesian Causal Forests (BCF) (Hahn et al., 2020) handles confounding and overlap issues natively by isolating the prognostic effect from the treatment effect, avoiding the extrapolation bias of standard BART models (Zhu et al., 2023).

3 The ZIC-BCF-Smear Estimator

The Zero-Inflated Continuous Bayesian Causal Forest with Duan’s Smearing (ZIC-BCF-Smear) model is a two-part Bayesian forest framework designed to resolve the bias-variance trade-offs of semicontinuous outcomes. It decouples the process into a binary participation stage (Probit BCF) and a positive continuous intensity stage (Gaussian BCF), fit strictly on active units. To resolve selection bias and log-scale misspecification bias, Subpopulation Propensity Adjustment (SPA) and Duan’s non-parametric smearing re-transformation are incorporated.

3.1 The Bayesian Causal Forest Parameterization

Because both stages of ZIC-BCF-Smear are Bayesian Causal Forests, the BCF parameterization of Hahn et al. (2020) is stated first. For a scalar response W_i , a binary treatment Z_i , and covariates X_i , BCF writes the conditional mean as the sum of two separate sum-of-trees ensembles,

$$E[W_i | X_i, Z_i] = \mu(X_i, \hat{\pi}_i) + Z_i \tau(X_i), \quad (3.1)$$

where $\mu(\cdot)$ is the *prognostic function*, giving the mean response under control, and $\tau(\cdot)$ is the *treatment function*, giving the conditional average treatment effect on the scale of W_i . Each of the two functions is given its own Bayesian Additive Regression Trees (BART) prior (Chipman et al., 2010), that is, its own independent ensemble of regularized trees with its own depth prior and its own leaf-scale hyperparameters, so that the prior on the treatment effect is chosen separately from and typically much tighter than the prior on the prognostic surface. The two ensembles are not fit in isolation: they are updated inside a *single* Gibbs sampler by Bayesian backfitting, each ensemble being updated against the partial residual left by the other, so that μ and τ are jointly identified from a common chain. The estimated propensity score $\hat{\pi}_i = \hat{P}(Z_i = 1 | X_i)$ is supplied as an additional input to the prognostic ensemble only; this is the device by which BCF avoids regularization-induced confounding, since a tree ensemble that is free to shrink the prognostic surface toward zero would otherwise absorb targeted selection into $\tau(\cdot)$ (Hahn et al., 2020).

ZIC-BCF-Smear instantiates (3.1) twice, once on a latent participation scale and once on the log-intensity scale. The subscript b denotes the binary hurdle stage and the subscript c the continuous intensity stage, so that μ_b and τ_b are the prognostic and treatment functions for participation, and μ_c and τ_c are the prognostic and treatment

functions for log intensity. Throughout, the abbreviations

$$\mu_{bi} \equiv \mu_b(X_i, \hat{\pi}_i), \quad \tau_{bi} \equiv \tau_b(X_i), \quad \mu_{ci} \equiv \mu_c(X_i, \hat{\pi}_i^+), \quad \tau_{ci} \equiv \tau_c(X_i) \quad (3.2)$$

are used for the four unit-level evaluations, and a superscript (s) denotes the value of a quantity at the s -th retained MCMC draw. Under this notation, μ_{ci} is the control log-intensity of unit i and $\mu_{ci} + \tau_{ci}$ is its treated log-intensity.

3.2 Stage 1: Hurdle Stage (Probit BCF)

The hurdle stage models the probability that an observation clears the zero threshold. Let $I_i = \mathbb{I}(Y_i > 0) \in \{0, 1\}$ be the participation indicator. This participation indicator I_i is represented using a latent Gaussian utility variable:

$$I_i^* = \mu_b(X_i, \hat{\pi}_i) + Z_i \tau_b(X_i) + \eta_i, \quad \eta_i \sim \mathcal{N}(0, 1) \quad (3.3)$$

where $I_i = \mathbb{I}(I_i^* > 0)$. The probability of clearing the hurdle is:

$$P(I_i = 1 \mid X_i, Z_i) = \Phi(\mu_b(X_i, \hat{\pi}_i) + Z_i \tau_b(X_i)) \quad (3.4)$$

Here $\Phi(\cdot)$ is the standard normal cumulative distribution function, $\mu_b(\cdot)$ is the prognostic hurdle forest modeled via Bayesian Additive Regression Trees (Chipman et al., 2010; Hill, 2011) and representing prognostic effects on participation, $\tau_b(\cdot)$ is the treatment-moderating hurdle forest representing treatment effect heterogeneity on participation, and $\hat{\pi}_i$ is the full-sample propensity score, included in $\mu_b(\cdot)$ to enforce close-up covariate adjustment and avoid regularizing away prognostic confounding (Hahn et al., 2020). The Probit BCF model is fit on the full sample of size N using data augmentation for the latent variables I_i^* . It is not a standalone fit: it is the first of two parameter blocks updated within each sweep of a single Gibbs sampler, whose second block is the intensity stage of Section 3.3.

3.3 Stage 2: Continuous Intensity Stage (Gaussian BCF)

For observations that clear the hurdle ($Y_i > 0$), the continuous response intensity Y_i^+ is modeled on the log scale to accommodate right-skewness:

$$\log(Y_i^+) = \mu_c(X_i, \hat{\pi}_i^+) + Z_i \tau_c(X_i) + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, \sigma_c^2) \quad (3.5)$$

Here $\mu_c(\cdot)$ is the prognostic continuous intensity forest modeling the baseline intensity, $\tau_c(\cdot)$ is the treatment-moderating intensity forest capturing heterogeneous treatment effects on the log scale, $\hat{\pi}_i^+ = P(Z_i = 1 \mid X_i, Y_i > 0)$ is the subpopulation propensity score estimated strictly on the active subset, and ϵ_i represents log-scale homoskedastic Gaussian errors with variance σ_c^2 . This continuous stage BCF is fit strictly on the active subset of size $N_{\text{active}} = \sum_i I_i$, which completely avoids the latent data augmentation noise typical of joint selection models.

Restricting the intensity block to the active units does not amount to fitting a separate model. Within each sweep the sampler updates the hurdle block on all N

units and then this block on the N_{active} units, and after burn-in writes both to the *same* draw index s , so every response-scale quantity in Section 3.5 is assembled from hurdle and intensity parameters belonging to one state of one chain. Nothing is lost by updating the blocks in turn rather than jointly: the two-part likelihood factorizes exactly into a participation factor over all N units and an intensity factor over the active units, the indicators $\{I_i\}$ being a deterministic function of the observed data, and the two blocks share no parameter. With independent BART priors on $\theta_b = (\mu_b, \tau_b)$ and $\theta_c = (\mu_c, \tau_c, \sigma_c^2)$ this yields $p(\theta_b, \theta_c \mid \text{data}) = p(\theta_b \mid \text{data}) p(\theta_c \mid \text{data})$, so the block-wise sweep is an exact Gibbs sampler for the joint posterior. The argument is given in full in Section 2 of the Supplementary Material (Souto et al., 2026).

3.4 Subpopulation Propensity Adjustment (SPA)

Conditioning on a post-treatment event ($Y_i > 0$, or $I_i = 1$) introduces selection bias (collider bias), since participation is itself affected by treatment and covariates. Under the full-sample strong ignorability assumption, potential outcomes are independent of treatment conditional on X_i . However, this independence does not automatically carry over to the subpopulation of active units:

$$(Y_i(0), Y_i(1)) \perp\!\!\!\perp Z_i \mid X_i \not\equiv (Y_i^+(0), Y_i^+(1)) \perp\!\!\!\perp Z_i \mid X_i, I_i = 1 \quad (3.6)$$

Assumption 3.1 (Subpopulation Strong Ignorability). Conditional on covariates X_i and active-status $I_i = 1$, the treatment assignment Z_i is independent of potential intensity outcomes:

$$(Y_i^+(0), Y_i^+(1)) \perp\!\!\!\perp Z_i \mid X_i, I_i = 1 \quad (3.7)$$

Remark 3.2. Assumption 3.1 is mathematically guaranteed to hold under two standard conditions: (i) full-sample strong ignorability, $(Y_i^+(0), Y_i^+(1), I_i(0), I_i(1)) \perp\!\!\!\perp Z_i \mid X_i$, and (ii) conditional two-part independence, $(Y_i^+(0), Y_i^+(1)) \perp\!\!\!\perp (I_i(0), I_i(1)) \mid X_i$. Under these conditions, the potential participation decisions do not carry additional information about the potential intensity outcomes beyond the covariates X_i , which prevents selection on unobservables (unobserved collider bias) and ensures that the distribution of potential continuous outcomes in the active subpopulation matches that in the full population.

Assumption 3.1, together with consistency and overlap within the active subpopulation, is what identifies the log-scale contrast that the continuous stage estimates.

Lemma 3.3 (Identification on the active subpopulation). *Let $L_i(z) = \log Y_i^+(z)$ and let $\tau_c^I(x) = E[L_i(1) - L_i(0) \mid X_i = x, I_i = 1]$. Suppose Assumption 3.1 holds, that $\log Y_i^+ = L_i(Z_i)$ whenever $I_i = 1$, and that $\pi^+(x) = P(Z_i = 1 \mid X_i = x, I_i = 1)$ is bounded away from 0 and 1 on the support of $X_i \mid I_i = 1$. Then*

$$\tau_c^I(x) = E[\log Y_i^+ \mid X_i = x, Z_i = 1, I_i = 1] - E[\log Y_i^+ \mid X_i = x, Z_i = 0, I_i = 1], \quad (3.8)$$

so $\tau_c^I(x)$ is identified from the active-subpopulation distribution, and the continuous-stage treatment function $\tau_c(x)$ represents this contrast.

The proof is given, with the formal statement of the overlap condition, in Section 2 of the Supplementary Material (Souto et al., 2026). One consequence deserves emphasis in the main text, because it is easy to over-read. The score $\pi^+(X_i)$ is a balancing score within the active subpopulation, so including it in $\mu_c(X_i, \pi^+(X_i))$ is a legitimate BCF regularization device that keeps subpopulation confounding out of $\tau_c(\cdot)$; but since $\pi^+(X_i)$ is itself a function of X_i , it adds no identifying power. It does not replace Assumption 3.1, active-subpopulation overlap, or adjustment for X_i when the target is heterogeneous in X_i .

3.5 Duan’s Non-Parametric Smearing Re-transformation

To obtain potential outcomes on the response scale, the log-scale predictions must be re-transformed back to the original scale. Under standard log-normal assumptions, the continuous expectation under treatment arm $z \in \{0, 1\}$ is:

$$E[Y_i^+(z) \mid X_i, I_i(z) = 1] = \exp(\mu_{ci} + z\tau_{ci}) \cdot \exp\left(\frac{\sigma_c^2}{2}\right) \quad (3.9)$$

with μ_{ci} and τ_{ci} as defined in (3.2), so that the treatment function enters through the term $z\tau_{ci}$ and drops out of the control arm. However, if the true positive outcome distribution is not strictly log-normal (e.g., Gamma or Tweedie), the parametric multiplier $\exp(\sigma_c^2/2)$ introduces severe misspecification scale bias. Under misspecification, outliers inflate the log-scale variance σ_c^2 , systematically scaling up all treatment effect estimates.

ZIC-BCF-Smear replaces the parametric multiplier with Duan’s Non-Parametric Smearing Estimator (Duan, 1983). Let $s \in \{1, \dots, S\}$ represent a saved MCMC draw:

1. **Compute active log-scale residuals:** For each active observation $j \in \{1, \dots, N_{\text{active}}\}$ where $Y_j > 0$, calculate the residual under the observed treatment Z_j :

$$e_j^{(s)} = \log(Y_j^+) - \left[\mu_c^{(s)}(X_j, \hat{\pi}_j^+) + Z_j \tau_c^{(s)}(X_j) \right] \quad (3.10)$$

2. **Calculate the smearing factor:** Compute the sample average of the exponentiated residuals:

$$\phi_{\text{smear}}^{(s)} = \frac{1}{N_{\text{active}}} \sum_{j \in \text{active}} \exp\left(e_j^{(s)}\right) \quad (3.11)$$

3. **Re-transform potential outcomes:** Reconstruct response-scale potential outcomes for all N units:

$$\mu_0^{(s)}(X_i) = \Phi\left(\mu_{bi}^{(s)}\right) \cdot \exp\left(\mu_{ci}^{(s)}\right) \cdot \phi_{\text{smear}}^{(s)} \quad (3.12)$$

$$\mu_1^{(s)}(X_i) = \Phi\left(\mu_{bi}^{(s)} + \tau_{bi}^{(s)}\right) \cdot \exp\left(\mu_{ci}^{(s)} + \tau_{ci}^{(s)}\right) \cdot \phi_{\text{smear}}^{(s)} \quad (3.13)$$

4. **Calculate raw CATE and ATE draws:**

$$\tau_{\text{resp}}^{(s)}(X_i) = \mu_1^{(s)}(X_i) - \mu_0^{(s)}(X_i) \quad (3.14)$$

$$\text{ATE}^{(s)} = \frac{1}{N} \sum_{i=1}^N \tau_{\text{resp}}^{(s)}(X_i) \quad (3.15)$$

3.6 Posterior Inference & Summaries

CATE and ATE point estimates are constructed as the posterior mean of the MCMC draws:

$$\hat{\tau}_{\text{resp}}(X_i) = \frac{1}{S} \sum_{s=1}^S \tau_{\text{resp}}^{(s)}(X_i), \quad \widehat{\text{ATE}} = \frac{1}{S} \sum_{s=1}^S \text{ATE}^{(s)} \quad (3.16)$$

Uncertainty quantification is provided by posterior credible intervals (CIs). The 95% credible interval for $\tau_{\text{resp}}(X_i)$ is bounded by the 2.5% and 97.5% quantiles of the posterior draws. The interval width is defined as the difference between these quantiles (credible interval length), and empirical coverage is defined as the proportion of units whose true CATE lies within their posterior interval.

These draw-level operations are well defined because $\mu_{bi}^{(s)}$ and $\mu_{ci}^{(s)}$ come from the same state of the same chain, as described in Sections 3.2 and 3.3. The one place where an approximation enters is the propensity scores: $\hat{\pi}_i$ and $\hat{\pi}_i^+$ are estimated once by logistic regression before the sampler starts and held fixed as covariates, so their estimation uncertainty is not propagated into the outcome posterior. This is a deliberate cut, and the same one made by BCF itself (Hahn et al., 2020), since feeding the propensity score into a joint posterior would let the flexible outcome model distort the treatment model. Its cost is recorded in Section 6. The complete fitting algorithm, the exactness argument for the block-wise sweep, the tree-prior hyperparameters, and the convergence diagnostics are given in Section 2 of the Supplementary Material (Souto et al., 2026).

3.7 Relation to Existing Models

ZIC-BCF-Smear combines the structural benefits of two-part hurdle models with the distribution-free flexibility of nonparametric causal forests. In Table 1, the proposed model is compared against existing semicontinuous causal estimators on key qualitative and structural characteristics.

Table 1: Qualitative comparison of semicontinuous causal estimators.

Feature	BCF-Linear	DPglm	EDP+ZI	Gamma Hurdle	Gamma + .01	ZIC-BCF-Smear
Separates Zero-Inflation	No	Yes	Yes	Yes	No	Yes
Hurdle CATE Estimation	No	Yes	Yes	Yes	No	Yes
MCMC Augmentation Noise	None	Extreme	Extreme	None	None	None
Subpopulation Confounding	N/A	Adjusted	Adjusted	Adjusted	N/A	Adjusted (SPA)
Scale Re-transformation	None	Parametric	Parametric	Parametric	Parametric	Non-Parametric

BCF-Linear ignores the hurdle process completely, fitting a single prognostic surface over the zero discontinuity, which produces large point error under right skewness. DPglm and EDP+ZI impute latent continuous values for all zero observations, causing extreme data augmentation noise and posterior variance inflation (DPglm’s CATE interval length ≈ 51.2). The parametric benchmarks are competitive under correct specification but suffer numerical instability and negative CATE correlation once link-linearity fails. ZIC-BCF-Smear decouples the hurdle from the continuous stage, adjusts for subpopulation confounding, and remains distribution-free via non-parametric smearing.

4 Simulation Studies

To evaluate the empirical performance of the proposed Zero-Inflated Continuous Bayesian Causal Forest with Duan’s Smearing (ZIC-BCF-Smear), a series of comprehensive simulation studies is conducted. In this section, the simulation design is outlined, the mathematical details of the data-generating processes (DGPs) under both link-linear and link-nonlinear specifications are presented, and the comparative results are analyzed.

4.1 Simulation Design, Estimators, and MCMC Settings

Five distinct estimators that represent different structural and distributional modeling assumptions are evaluated and compared:

1. **BCF-Linear (Single-Part Gaussian):** A standard continuous BCF model ([Hahn et al., 2020](#)) fit directly on the raw semicontinuous outcome Y_i , ignoring the discrete point mass at zero.
2. **DPglm (Bayesian Nonparametric):** A Dirichlet-process mixture of zero-inflated generalized linear models ([Oganisian et al., 2021](#)). This model cluster-imputes latent continuous values for all zero observations at each iteration, representing a flexible Bayesian nonparametric benchmark.
3. **Gamma Hurdle (Parametric Two-Part):** A standard two-part model comprising a logistic regression hurdle and a Gamma log-link continuous intensity regression ([Oganisian et al., 2021](#)), fit as weakly informative Bayesian GLMs with a fully treatment-interacted design and combined via G-computation.
4. **Gamma + .01 (Naive Single-Part):** A naive single-part Gamma regression model with a log link, fit to the outcome after adding a small positive constant ($Y_i + 0.01$) to resolve the zero outcomes ([Oganisian et al., 2021](#)).
5. **ZIC-BCF-Smear (Proposed):** The proposed two-part BCF model that decouples the probit hurdle and active continuous intensity stage, utilizing Duan’s non-parametric smearing to reconstruct response-scale effects.

For all BCF-based models, MCMC settings are standardized to 1,000 burn-in iterations followed by 1,000 post-warmup sampling iterations, which was found to be sufficient for posterior convergence. The performance of each estimator is evaluated across 100 independent Monte Carlo replications (seeds) per cell. Performance is assessed using six key metrics: CATE Root Mean Squared Error (RMSE), CATE absolute bias, CATE 95% credible interval coverage, CATE correlation (the Pearson correlation between estimated and true CATEs), CATE credible interval length, and population-level ATE RMSE and absolute bias.

Why EDP+ZI is absent from the benchmark. A sixth estimator, the enriched Dirichlet process zero-inflated model (EDP+ZI) of [Kim et al. \(2024\)](#), was implemented and

verified against the authors' reference code but could not be carried into the simulation grid. Its Gibbs updates assign observations to latent clusters through multivariate normal marginal-likelihood evaluations and joint covariance updates that scale cubically in the sample size, $O(N^3)$, whereas a BCF forest costs $O(T \cdot P \cdot N \log N)$ per sweep for T trees and P covariates. The empirical consequence is stark: at $N = 500$ a single standard fit takes ≈ 18 minutes for ZIC-BCF-Smear against ≈ 27.5 hours for EDP+ZI, a $\approx 90\times$ slowdown, and the cubic term pushes a single EDP+ZI fit past nine days at $N = 1,000$. Across the full design (3 DGPs, 4 sample sizes, 4 zero-inflation levels, 2 link regimes, 100 seeds) it would require on the order of 10^5 CPU hours, roughly a decade of continuous single-machine runtime, against capped academic computing sessions. It was therefore excluded, and that absence is recorded as a limitation in Section 6. The exclusion is itself a substantive result: an estimator that cannot be run 100 times on a 500-observation problem cannot be validated by simulation at all, which is precisely the barrier the forest-based formulation removes. The full timing analysis is given in Section 3 of the Supplementary Material (Souto et al., 2026).

4.2 Data-Generating Processes (DGPs)

For all DGPs, $P = 5$ baseline covariates are generated independently as $X_{ip} \sim \mathcal{N}(0, 1)$, and confounding is introduced through a treatment propensity score that is non-linear in X_{i2} :

$$\pi(X_i) = P(Z_i = 1 \mid X_i) = \Phi(-0.5 + 0.4X_{i1} + 0.3X_{i2}^2), \quad Z_i \sim \text{Bernoulli}(\pi(X_i)). \quad (4.1)$$

Outcomes are then generated under three distribution families, each in a link-linear and a link-nonlinear version, giving six DGPs. Throughout, c is the intercept that calibrates the marginal zero proportion, fixed at $\approx 40\%$ in the standard configuration and swept from 3% to 85% in Section 4.6.

Link-linear DGPs. DGP A and B share the probit participation stage

$$I_i \sim \text{Bernoulli}(\Phi(c + 0.5X_{i1} - 0.3X_{i3} + Z_i(0.2 + 0.1X_{i1}))), \quad (4.2)$$

and the common log-intensity predictor $\ell(X_i, Z_i) = 1.5 + 0.8X_{i2} + 0.4X_{i4} + Z_i(0.25 - 0.15X_{i2})$, with $Y_i = I_i \cdot Y_i^+$. In DGP A the positive part is log-normal, $\log(Y_i^+) \sim \mathcal{N}(\ell(X_i, Z_i), \sigma^2)$ with $\sigma = 0.5$, so the log-link continuous stage of ZIC-BCF-Smear is nearly correctly specified; in DGP B it is Gamma, $Y_i^+ \sim \text{Gamma}(\text{shape} = 2.0, \text{scale} = \exp\{\ell(X_i, Z_i)\}/2.0)$, so the parametric Gamma Hurdle benchmark is correctly specified and ZIC-BCF-Smear is not. Setting $c = 0.2$ gives $\approx 37\%$ and $\approx 39.5\%$ zeros respectively. DGP C instead draws the outcome directly from an intrinsic compound Poisson-Gamma dispersion model, $Y_i \sim \text{Tweedie}(\mu_i, \phi = 1.5, p = 1.5)$ with

$$\log \mu_i = 1.2 + c + 0.8X_{i1} - 0.4X_{i3} + Z_i(0.3 + 0.15X_{i1}), \quad (4.3)$$

and $c = 0$ giving $\approx 50\%$ zeros. Because the zero mass and the positive skewness are governed jointly by a single log-mean, no two-part partition exists and *every* hurdle-based estimator in the comparison, including the proposed one, is structurally misspecified. DGP C is included precisely to measure that cost.

Link-nonlinear DGPs. The affine link-scale predictors are replaced by mean-centered sinusoids, cross terms, and bounded treatment-effect functions built from four shared blocks:

$$h(X_i) = 0.50 \sin(1.3X_{i1}) - 0.30X_{i3} + 0.35X_{i1}X_{i2} - 0.20(X_{i3}^2 - 1), \quad (4.4)$$

$$\tau_h(X_i) = 0.20 + 0.20X_{i1}X_{i2}, \quad (4.5)$$

$$m(X_i) = 1.50 + 0.80 \sin(1.2X_{i2}) + 0.40X_{i4} + 0.35X_{i2}X_{i4} - 0.25(X_{i1}^2 - 1), \quad (4.6)$$

$$\tau_c(X_i) = 0.25 - 0.15 \sin(1.3X_{i2}) + 0.12 \tanh(X_{i1}X_{i2}). \quad (4.7)$$

The quadratic and sinusoidal terms are centered to have expectation ≈ 0 under $X \sim \mathcal{N}(0, I_5)$, which holds the outcome scale and the zero proportion comparable to the linear settings so that the two regimes differ in functional form alone. Nonlinear A and B replace (4.2) by participation probabilities $p_z(X_i) = \Phi(c + h(X_i) + z\tau_h(X_i))$ and replace $\ell(X_i, Z_i)$ by $m(X_i) + Z_i\tau_c(X_i)$, keeping their respective log-normal and Gamma positive parts; nonlinear C keeps $\phi = 1.5$ and $p = 1.5$ and sets $\log \mu_i = c + g_c(X_i) + Z_i\tau_{\text{Tweedie}}(X_i)$ with $g_c(X_i) = 1.2 + 0.80 \sin(1.2X_{i1}) - 0.40X_{i3} + 0.35X_{i1}X_{i3} - 0.25(X_{i2}^2 - 1)$ and $\tau_{\text{Tweedie}}(X_i) = 0.30 - 0.15 \sin(1.3X_{i1}) + 0.12 \tanh(X_{i1}X_{i3})$. The intercepts $c = 0.2$ (A, B) and $c = -2.072$ (C) calibrate all three to $\approx 40\%$ zeros.

The link-linear regime is the classical benchmark design and the one most favorable to the parametric competitors, since their fixed $(1, z, X, z \cdot X)$ design then matches the truth exactly. The link-nonlinear regime leaves the forests correctly flexible while making the parametric benchmarks wrong in functional form but not in distributional family, which isolates functional misspecification as the single active ingredient. The closed-form true potential outcomes $\mu_0(X_i)$ and $\mu_1(X_i)$ against which every estimator is scored, together with the calibrating intercepts for each zero-inflation level, are given in Section 1 of the Supplementary Material (Souto et al., 2026).

4.3 Link-Linear DGP Comparative Results

Performance is first analyzed under the standard configuration ($N = 500$, nominal zero-inflation, 100 seeds) where conditional-mean surfaces are linear on the link scale. Under this setup, the parametric Gamma Hurdle model is correctly specified for DGP B and nearly correctly specified for DGP A. The comparative results are summarized in Table 2.

Under hurdle DGPs (A and B), ZIC-BCF-Smear achieves the lowest CATE RMSE (1.526 and 1.380) and the narrowest credible interval lengths, demonstrating that the proposed decoupled forest structure is highly precise. Crucially, ZIC-BCF-Smear matches or outperforms the parametric Gamma Hurdle model (1.810 and 1.906) even though the latter is correctly specified for DGP B and nearly correctly specified for DGP A. This shows that committing to a parametric distribution for the positive part can be avoided without sacrificing estimation precision.

In contrast, BCF-Linear performs poorly under hurdle right-skewness (RMSE 3.662 and 4.010) because it forces a single Gaussian fit over the zero discontinuity. Naive Gamma + .01 is similarly biased, resulting in high CATE RMSE (2.946 and 3.076)

Table 2: Performance metrics compiled across 100 independent seeds at the standard configuration ($N = 500$, ZI Level 3, $\approx 40\%$ zeros) under link-linear DGPs.

DGP	Model	CATE RMSE	CATE Cov	CATE Corr	CATE CI Len	ATE RMSE	ATE Abs Bias
DGP A: Log-Normal	BCF-Linear	3.662	93.2%	0.184	6.315	0.849	0.282
	DPglm	1.409	100.0%	0.321	51.210	0.781	0.637
	Gamma Hurdle	1.810	94.9%	0.419	4.641	0.585	0.029
	Gamma +.01	2.946	91.7%	0.267	6.215	0.765	0.212
	ZIC-BCF-Smear	1.526	94.9%	0.397	3.819	0.614	0.406
DGP B: Gamma	BCF-Linear	4.010	92.5%	0.156	6.489	0.812	0.253
	DPglm	1.406	100.0%	0.275	51.759	0.713	0.520
	Gamma Hurdle	1.906	93.7%	0.370	4.860	0.629	0.024
	Gamma +.01	3.076	91.1%	0.259	6.350	0.836	0.246
	ZIC-BCF-Smear	1.380	95.7%	0.398	4.174	0.650	0.339
DGP C: Tweedie	BCF-Linear	5.387	85.3%	0.723	7.379	0.715	0.169
	DPglm	4.837	99.9%	0.267	51.351	1.003	0.793
	Gamma Hurdle	2.067	95.0%	0.944	4.961	0.534	0.051
	Gamma +.01	2.549	95.1%	0.914	5.662	0.614	0.012
	ZIC-BCF-Smear	2.471	98.2%	0.916	7.404	0.850	0.224

and sub-nominal coverage (91.1%–91.7%), which highlights the danger of shifting zero outcomes arbitrarily. While DPglm attains low CATE RMSE, its posterior credible intervals are pathologically over-wide (≈ 51.2 compared to ZIC-BCF-Smear’s 3.8), pinning coverage at a degenerate 100% and rendering it unusable for calibrated uncertainty quantification. That interval inflation is the direct signature of the decoupling argument of Section 3.3: because DPglm fits its continuous stage on the full sample by imputing a latent continuous value for every zero at every sweep, roughly 40% of the continuous data it splits on at the standard configuration is simulated MCMC noise, which propagates into the cluster and split proposals and inflates the posterior. ZIC-BCF-Smear trains its intensity forest only on units with $Y_i > 0$ and therefore never pays that price, which is why it can be simultaneously more accurate and an order of magnitude sharper.

In DGP C, which features an intrinsic compound Tweedie structure without a hurdle partition, the parametric Gamma Hurdle is the strongest single model; however, ZIC-BCF-Smear remains competitive, achieving low RMSE (2.471) and positive correlation (0.916). This is the expected cost of imposing a two-part factorization on a process that does not have one, and it bounds the penalty: even where the structural assumption of the method is false by construction, it stays within 20% of the best model on CATE RMSE and retains a correlation above 0.9.

4.4 Link-Nonlinear DGP Comparative Results

Performance is next evaluated under link-nonlinear conditional-mean surfaces. Under this setup, the fixed linear design $(1, z, X, z \cdot X)$ of the parametric benchmarks (Gamma Hurdle and Gamma + .01) is misspecified. Standard cell results are presented in Table 3; the accompanying numerical-stability metrics are reported in Section 4 of the Supplementary Material (Souto et al., 2026).

Under link-nonlinearity, the parametric Gamma benchmarks break down in two ways. First, they fail to capture treatment heterogeneity, yielding negative CATE correlations in all three DGPs (ranging from -0.039 to -0.081). This shows that a misspecified linear model ranks treatment effects in roughly the wrong order, which for a

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Table 3: Standard comparative results ($N = 500$, ZI Level 3, $\approx 40\%$ zeros) under link-nonlinear DGPs. Dagger ($\dagger$) indicates metrics contaminated by numerical divergence.

DGP	Model	CATE RMSE	CATE Cov	CATE Corr	CATE CI Len	ATE RMSE	ATE Abs Bias
DGP A: Log-Normal	BCF-Linear	5.812	88.1%	0.347	6.385	1.127	0.172
	DPglm	3.070	99.7%	0.198	61.532	0.680	0.021
	Gamma Hurdle	$9.7 \times 10^7 \dagger$	69.2%	-0.039	$5.8 \times 10^7 \dagger$	$5.6 \times 10^8 \dagger$	$5.9 \times 10^7 \dagger$
	Gamma +.01	$3.0 \times 10^9 \dagger$	69.5%	-0.081	$2.1 \times 10^{11} \dagger$	$7.5 \times 10^8 \dagger$	$7.3 \times 10^7 \dagger$
	ZIC-BCF-Smear	1.734	89.7%	0.686	4.422	0.537	0.119
DGP B: Gamma	BCF-Linear	5.792	89.1%	0.318	6.624	1.101	0.217
	DPglm	2.639	99.5%	0.211	61.457	0.716	0.126
	Gamma Hurdle	$1.7 \times 10^{10} \dagger$	74.1%	-0.040	$1.5 \times 10^{11} \dagger$	$7.5 \times 10^{10} \dagger$	$1.1 \times 10^{10} \dagger$
	Gamma +.01	$5.1 \times 10^8 \dagger$	68.9%	-0.068	$9.7 \times 10^8 \dagger$	$8.7 \times 10^8 \dagger$	$5.0 \times 10^7 \dagger$
	ZIC-BCF-Smear	1.656	91.7%	0.645	4.753	0.641	0.067
DGP C: Tweedie	BCF-Linear	0.302	91.6%	0.060	0.638	0.111	0.026
	DPglm	0.404	99.9%	0.022	37.874	0.114	0.058
	Gamma Hurdle	$3.8 \times 10^{10} \dagger$	81.4%	-0.060	$1.7 \times 10^{11} \dagger$	$6.3 \times 10^{10} \dagger$	$7.8 \times 10^9 \dagger$
	Gamma +.01	$1.9 \times 10^{10} \dagger$	76.3%	-0.045	$2.2 \times 10^{11} \dagger$	$1.2 \times 10^{11} \dagger$	$1.2 \times 10^{10} \dagger$
	ZIC-BCF-Smear	0.250	94.6%	0.175	0.704	0.101	0.030

CATE learner is a more serious failure than a scale error: a decision rule that targets treatment to the units with the largest estimated benefit would, under these benchmarks, select approximately the wrong units. Second, the misspecified linear predictors generate extreme fitted values on tail covariate combinations, causing the link-exp($\cdot$) and penalized-IRLS updates to overflow. Divergence occurs on 4 to 5 of 100 seeds for Gamma Hurdle and 5 to 9 for Gamma + .01, inflating the mean CATE RMSE and ATE metrics to the 10^7 – 10^{11} range. Robust medians computed after excluding the diverged seeds are reported in Section 4 of the Supplementary Material (Souto et al., 2026); they do not rescue the benchmarks, since the negative CATE correlations persist and the median CATE RMSEs remain at 3.7 to 4.3 in DGP A and B.

The mechanism behind that overflow is exactly what Duan’s smearing is introduced to defuse. A parametric log-scale model re-transforms through a multiplier fixed by its assumed family, $\exp(\sigma_c^2/2)$ in the log-normal case; when the model is wrong in functional form, the residuals it fails to explain inflate σ_c^2 , the multiplier grows exponentially in that inflation, and the error compounds on precisely the tail covariate combinations where the linear predictor is already most extreme. The smearing factor $\phi_{\text{smear}}^{(s)}$ of Section 3.5 replaces it with the empirical mean of exponentiated residuals, which is bounded by the observed data. ZIC-BCF-Smear is therefore not merely more accurate here but categorically more stable, recording zero diverged seeds across all 100 replications of all three nonlinear DGPs.

Correspondingly, ZIC-BCF-Smear achieves the lowest CATE RMSE (1.734 / 1.656 / 0.250) across all DGPs, outperforming the robust medians of the parametric models by a factor of 2.0 to 2.7. It also maintains positive CATE correlations (0.686 / 0.645 / 0.175), successfully identifying treatment effect direction and magnitude. Under nonlinearity, the ATE “refuge” that Gamma Hurdle enjoyed in the linear study collapses: ZIC-BCF-Smear yields the lowest ATE RMSE in all three DGPs. BCF-Linear is stable but weak under hurdle skew (RMSE ≈ 5.8 , coverage $\approx 88\%$ in A/B), while DPglm remains pathologically over-wide (CI length ≈ 61). The one blemish is a mild under-coverage of ZIC-BCF-Smear in DGP A (89.7%) and DGP B (91.7%), traced in Section 6 to the conservative BART regularization priors, which shrink the posterior toward the prior

mean for units in the tails of the covariate distribution.

4.5 Hurdle Stage Performance

Because the response-scale effect is a product of a participation probability and an intensity, the quality of the hurdle stage is as consequential as that of the intensity stage, and it is separately identified. Estimators that decouple zero-inflation from positive outcomes model the binary indicator $I_i = \mathbb{I}(Y_i > 0)$ in the first stage, so the hurdle metrics of ZIC-BCF-Smear (Probit BCF), DPglm (Dirichlet process mixture), and Gamma Hurdle (logistic GLM) can be compared directly; BCF-Linear and Gamma + .01 do not model a hurdle and are absent from this comparison. Table 4 reports the standard configuration under both link regimes.

Table 4: Hurdle-stage performance at the standard configuration ($N = 500$, ZI Level 3, $\approx 40\%$ zeros), across 100 seeds. RMSE is on the participation-probability scale and coverage is the empirical 95% credible-interval coverage of the true participation probability. BCF-Linear and Gamma + .01 do not model a hurdle stage.

DGP	Model	Link-Linear		Link-Nonlinear	
		Hurdle RMSE	Coverage	Hurdle RMSE	Coverage
DGP A: Log-Normal	DPglm	0.053	99.7%	0.077	98.6%
	Gamma Hurdle	0.098	94.0%	0.129	88.6%
	ZIC-BCF-Smear	0.046	99.3%	0.065	94.7%
DGP B: Gamma	DPglm	0.055	99.7%	0.077	98.2%
	Gamma Hurdle	0.098	94.0%	0.129	88.5%
	ZIC-BCF-Smear	0.046	99.3%	0.066	94.5%
DGP C: Tweedie	DPglm	0.044	87.0%	0.057	99.8%
	Gamma Hurdle	0.081	95.0%	0.111	92.4%
	ZIC-BCF-Smear	0.028	99.3%	0.044	98.4%

ZIC-BCF-Smear attains the lowest hurdle RMSE in every cell of the table, under both link regimes and all three DGPs. Under link-linearity its advantage over the logistic GLM of the Gamma Hurdle benchmark (0.046 versus 0.098 in DGP A and B) has a clean explanation: the true participation stage in (4.2) is a probit, so the Probit BCF stage matches the true link while the logistic GLM is mildly misspecified even in the regime designed to favor it. Under link-nonlinearity the logistic GLM is misspecified in functional form as well, its RMSE rises to 0.129, and its coverage collapses to $\approx 88.5\%$; ZIC-BCF-Smear holds RMSE at 0.065–0.066 with essentially nominal coverage (94.5%–94.7%). DPglm is accurate but uninformative, its hurdle credible intervals averaging a length of ≈ 1.99 on a probability scale bounded in $[0, 1]$, which is a second manifestation of the augmentation-noise inflation seen on the response scale. Full hurdle-stage results across sample sizes and zero-inflation levels are tabulated in Section 4 of the Supplementary Material (Souto et al., 2026).

4.6 Sensitivity to Sample Size and Zero-Inflation

Two sensitivity sweeps are run: a sample-size grid $N \in \{100, 250, 500, 1000\}$ under nominal zero-inflation, and a zero-inflation grid spanning realized zero proportions from 3% to 85% at $N = 500$. Both are run under both link regimes and all three DGPs, giving 48 additional cells. The complete numerical results and all sensitivity figures are reported in Section 4 of the Supplementary Material ([Souto et al., 2026](#)); the substantive findings are four.

First, ZIC-BCF-Smear is asymptotically well behaved. Under link-linearity its CATE RMSE falls and its credible intervals contract systematically with N (interval length $6.73 \rightarrow 4.82 \rightarrow 3.82 \rightarrow 3.20$ across the grid in DGP A) while coverage holds at the nominal level, as does the correctly specified Gamma Hurdle. DPglm’s intervals contract too but remain an order of magnitude wider ($99.19 \rightarrow 42.61$), so its coverage stays pinned near 100% at every sample size; BCF-Linear contracts but its coverage collapses (to 83.5% under Tweedie at $N = 1,000$), and Gamma + .01 diverges at $N = 100$.

Second, the parametric failure under nonlinearity is not a small-sample artifact and does not wash out with data. The parametric benchmarks diverge in 23 of the 24 nonlinear sensitivity cells, with divergence rates up to 10%, and their CATE correlations remain negative at every sample size, including $N = 1,000$. Misspecification in functional form is not something more observations repair.

Third, the advantage of the decoupled structure scales directly with the amount of zero-inflation, which is the design claim of the method. At $\approx 85\%$ zeros the CATE RMSE of ZIC-BCF-Smear in DGP A is 17.9% (link-linear) and 27.5% (link-nonlinear) of the single-part BCF-Linear baseline. The reason is structural rather than incidental: BCF-Linear must resolve the zero mass with the same smooth surface it uses for the positive part, and the more mass sits at zero the more that single surface is dominated by the discontinuity, whereas the ZIC-BCF-Smear intensity forest is never asked to see a zero at all.

Fourth, extreme zero-inflation is where the competitors break rather than merely degrade. At 85% zeros in DGP B the correctly specified parametric hurdle collapses to a normalized RMSE of 1.372, worse than the BCF-Linear baseline, because with only $\approx 15\%$ positive observations the Gamma GLM has too little active data to fit; Gamma + .01 diverges outright; and DPglm collapses under Tweedie (normalized RMSE 14.424), its latent imputation being almost pure noise when the active fraction is small. ZIC-BCF-Smear remains stable across the entire range and outperforms every competitor at the extremes of both grids.

5 Applied Study: Severe Adverse-Event Burden in a Head-and-Neck Cancer Trial

The simulation studies establish how ZIC-BCF-Smear behaves when the truth is known. This section applies it where it is not, to individual-participant data from SPECTRUM, an open-label phase III randomized trial of cisplatin and fluorouracil with or without

panitumumab in recurrent or metastatic squamous-cell carcinoma of the head and neck (Vermorken et al., 2013; ClinicalTrials.gov, 2026), obtained through Project Data Sphere (Project Data Sphere, 2026). A condensed account is given here; the full study, with the complete subgroup tables, the exposure and missingness diagnostics, the model comparison, the literature positioning, and the reproducibility trail, is Section 5 of the Supplementary Material (Souto et al., 2026).

5.1 Data, Outcome, and Estimand

The analytic extract contains 520 participants balanced at 260 per arm, with $Z_i = 1$ denoting panitumumab plus chemotherapy. Because the trial randomized assignment, the unadjusted arm difference is already unbiased for the average effect in this cohort; the baseline adjustment set is a variance-reduction device rather than a requirement for validity, and the known design propensity is 0.5. That guarantee attaches to the randomization, so it holds for the 520-person extract to the extent that the extract is a randomization-respecting subset of the parent trial’s 657 randomized patients. The supplied files do not record how the 657 became 520, and the exactly balanced 260 per arm cannot by itself distinguish a benign completeness filter from a selection that differs by arm. Section 5 of the Supplementary Material states the reconciliation that is checkable from the supplied files and flags the step that is not.

The covariate vector X_i contains age, ECOG performance status, sex, diagnosis site, HPV status, disease status, and the patient-level prior-radiotherapy indicator PRRADIO. Prior radiotherapy is present in 76.9% of the chemotherapy arm and 83.1% of the panitumumab arm and is included because it is clinically relevant to the mucosal and cutaneous toxicity that the outcome measures. The broader composite prior-treatment variable PRHNTRTC is reported descriptively but is not entered alongside PRRADIO, in the adjustment or the subgroup lists, because the two indicators differ for only four patients and their near-redundancy destabilized the Gamma benchmark while adding an unadjusted near-duplicate contrast to every subgroup grid. Tumor category is excluded because it is a deterministic recode of diagnosis site in this extract, so carrying both places an exact linear combination of existing columns in the design matrix.

The outcome is the patient-level burden of severe toxicity, $Y_i^{\text{sum}} = \sum_{j \in \mathcal{A}_i} d_{ij} \mathbf{1}(G_{ij} \geq 3)$, where $\mathcal{A}_i$ indexes the on-study adverse-event records for patient i , d_{ij} is the recorded duration of record j , and G_{ij} its recorded grade. This is exactly the regime the estimator targets: 123 patients (23.7%) accrue no severe-adverse-event time at all, while the positive part is heavily right-skewed with a median of 26 and a maximum of 641 event-days. Because Y_i^{sum} sums record durations rather than taking the union of calendar intervals, concurrent severe events contribute more than one event-day on the same date. A companion endpoint Y_i^{cal} , the number of distinct on-study calendar days on which at least one grade 3+ event was ongoing, is therefore also constructed by merging overlapping intervals within each patient; the union removes 27.1% of the record-days but leaves the zero count unchanged. Of the 1,671 qualifying severe records, 1,547 have a complete imputed interval and 124 do not, in every case because the end day rather than the start day is unresolved. Those 124 records fall across 88 patients, 51 on panitumumab and 37 on chemotherapy, whose calendar-day totals are correspondingly understated,

and a smaller group of 13 patients has no complete qualifying interval at all and is assigned zero outright. That group is imbalanced, 10 on panitumumab against 3 on chemotherapy, which runs in the direction of understating the panitumumab arm and lands on the hurdle component. The v2 disposition file supplies DSDY, the recorded safety-follow-up day, and LASTCTDY, the last-contact study day, for all 520 patients. Only 354 patients completed the safety follow-up, however, and three complete intervals end after LASTCTDY. Thus the new domains support an observed-window censoring diagnostic but do not by themselves establish a fully observed safety window; an unresolved end day still cannot be read as evidence that the event was ongoing at data collection.

For either outcome the target is the analytic-cohort average treatment effect $\Delta^k = \mathbb{E}\{Y_i^k(1) - Y_i^k(0) \mid i \in \mathcal{S}\}$ over the 520-person cohort $\mathcal{S}$, which under randomization and consistency is the effect of *assignment* on accrued severe-adverse-event days under the trial’s own differential treatment duration. Treatment duration is not held fixed and is not adjusted for, deliberately: panitumumab continues as maintenance after chemotherapy stops, so time on treatment is a post-randomization consequence of assignment and lies on the causal pathway from assignment to accrued burden. It can therefore be neither ignored in interpretation nor naively conditioned on in estimation. Section 5.3 quantifies how much of the effect travels through that channel. The estimand is not a fixed-horizon restricted mean, not a rate per unit of exposure, and not a competing-risk estimand.

5.2 Average Effects and Endpoint Sensitivity

ZIC-BCF-Smear was fit as four chains of 4,000 retained draws after 5,000 burn-in iterations, and the Bayesian Gamma hurdle model of [Oganisian et al. \(2021\)](#) was fit to the same cohort, covariates, and treatment coding as a specification benchmark, likewise as four chains. Across the global monitored functionals the largest Gelman–Rubin statistic is 1.001 and the smallest effective sample size is 2,670 for ZIC-BCF-Smear; the corresponding Gamma-hurdle values are 1.000 and 16,000 on the same four-chain, 4,000-draw-per-chain budget ([Gelman and Rubin, 1992](#); [Plummer et al., 2006](#)). The complete diagnostic tables are in the Supplementary Material. Table 5 reports both models on both endpoints, together with the participation margin.

Three results follow. First, the average effect is credibly positive and survives the change of outcome definition: ZIC-BCF-Smear estimates 14.27 additional event-days on the summed scale (posterior probability of a positive effect 0.999) and 10.00 additional days once concurrent events are counted once (0.999). Both the sign and the near-certainty agree with the parent trial’s qualitative safety finding, namely that the panitumumab-containing regimen produced more severe toxicity across several conventional adverse-event categories ([Vermorken et al., 2013](#)), and the conclusion survives the one outcome-definition change that could plausibly have manufactured it, the double-counting of concurrent severe events. The 4.27-day attenuation measures how much of the summed burden reflects simultaneity, not a reversal of the safety conclusion. What the literature cannot supply here is a numerical benchmark, since standard safety tables summarize event incidence, grade, and category rather than summed record durations

Table 5: Posterior average treatment effects of panitumumab assignment on severe-adverse-event burden, SPECTRUM analytic cohort ($N = 520$). Positive values indicate a higher burden, or a higher probability of a positive reconstructed severe-adverse-event burden, under panitumumab. The hurdle margin is not literally the probability of any severe adverse event: 13 patients with severe events are reconstructed as zeros because their observation intervals were incomplete. The first four rows are on the event-day scale; the last two are differences in probability. The unadjusted arm differences, which randomization makes unbiased for the same estimand, are 15.68 and 12.06 event-days.

Estimand	Model	Estimate	95% Credible Interval
Summed record duration (event-days)	ZIC-BCF-Smear	14.27	[4.69, 24.23]
	Gamma Hurdle	15.95	[6.15, 26.30]
Distinct calendar days	ZIC-BCF-Smear	10.00	[3.57, 16.62]
	Gamma Hurdle	12.20	[5.56, 19.23]
Participation margin (probability)	ZIC-BCF-Smear	0.029	[−0.038, 0.102]
	Gamma Hurdle	0.017	[−0.055, 0.089]

or overlap-adjusted calendar days ([Vermorken et al., 2013](#)); external agreement is therefore assessable for direction and robustness but not for magnitude, which is exactly the gap this endpoint fills.

Second, the two specifications agree on the average, at 15.95 and 12.20 days for the Gamma hurdle, with a gap between point estimates that is small relative to the width of either posterior, so the finding is not an artifact of either functional form. Both also sit close to the corresponding unadjusted arm differences of 15.68 and 12.06 event-days. Under randomization that proximity is the benchmark rather than a weakness: the unadjusted contrast is itself unbiased for the cohort average effect, so a model that departed sharply from it would need to explain why. What the adjusted fits add is precision and the ability to resolve the effect by covariate profile and by component, not a correction to a biased raw comparison.

Third, and most informative for the two-part structure, the participation margin is uncertain under both models. Since 76.3% of the cohort has a positive burden in both arms, that margin has little room to move, and the credible effect is carried mainly by the positive-intensity component. The supported claim therefore concerns how much severe-toxicity time accumulates among patients whose reconstructed burden is positive, rather than whether that reconstructed burden is positive at all. That attribution is a direct product of decoupling the two stages; a single-part estimator on the raw day scale returns the same aggregate number with no way to localize it, and the published safety evidence, which emphasizes particular event categories and grades rather than an aggregate any-severe-AE indicator ([Vermorken et al., 2013](#)), contains no quantity that either confirms or refutes this margin. The defensible interpretation is a difference concentrated in positive intensity, which can reflect event duration, event multiplicity, overlap, treatment exposure, and incomplete interval reconstruction rather than persistence alone. One qualification belongs with that reading, since the 13 patients forced to

zero by an unresolved interval boundary are imbalanced 10 to 3 against panitumumab and enter precisely this component.

5.3 Treatment Exposure and What the Estimand Does Not Cover

The endpoint counts days in a severe-adverse-event state, and that quantity accrues with time on treatment. Exposure is grossly unequal, and structurally so: mean treatment duration is 83.7 days in the chemotherapy arm against 130.3 days under panitumumab, a 56% difference, with maxima of 276 and 1,149 days. The consequence is large. Truncating every patient’s observed accrual at a common study day collapses the arm difference in distinct severe-adverse-event days from 12.06 days uncapped to 4.10 at 120 days, 3.37 at 90 days, and 3.14 at 84 days, and the same attenuation reproduces across all four start-date and end-date variants recorded in the file.

Two cautions govern how that table is read, and they matter more than the numbers. It is truncation of observed accrual, not a fixed-horizon estimand: capping at day 84 removes events recorded after day 84 but does not create observation for a patient whose follow-up ended earlier. The new disposition domains now make that distinction measurable. Using LASTCTDY as an observed upper bound gives mean uncapped observed-window burdens of 19.18 and 30.90 days and, at 84 days, 11.25 and 14.33 days in the chemotherapy and panitumumab arms, respectively; 47 and 41 patients had last contact before day 84. These are still diagnostics, not fixed-horizon causal estimates, because early contact loss and incomplete interval boundaries require a censoring strategy. Whether the effect reverses on a per-exposure basis also depends on the normalization, since the patient-level mean of per-100-treatment-day rates reverses, at 53.7 against 49.9, while the aggregate person-time rate does not, at 22.9 against 24.0. Fitting the same two models to the per-exposure rate reproduces that ambiguity rather than resolving it: ZIC-BCF-Smear returns 5.19 days per 100 treatment-days $[-4.39, 17.28]$ and the Gamma hurdle $-5.23 [-37.06, 25.19]$, both compatible with no effect. The defensible claim is severe attenuation once exposure is accounted for, not reversal. The horizon diagnostic provides the first check on that collapse, and it is deliberately weaker than a corrected estimate. Table 6 reports means among all patients and among patients whose last contact reaches each horizon. The arm difference is not smaller among patients observed through the horizon, going from 3.08 to 3.68 days at day 84 and from 4.03 to 5.13 days at day 120, so simple truncation of the shortest-observed records is not what produces the attenuation. Restricting to patients observed through the horizon conditions on a post-randomization variable, however, so informative censoring remains possible and the direction of any resulting selection bias cannot be signed in advance; these rows are therefore reported as diagnostics, not as a corrected fixed-horizon effect.

None of this is offered as a corrected causal estimate, and the randomized analysis of Section 5.2 is left standing rather than replaced. Treatment duration is measured after randomization, so dividing by it conditions on a post-treatment variable and can induce selection bias whose direction is not fixed in advance: a patient who stops early because of toxicity contributes a short denominator alongside a large numerator, inflating that patient’s rate and making their arm look worse, whereas a patient who stops early for an unrelated reason pushes the same arm’s rate down. Which dominates depends

Table 6: Mean accrued severe-adverse-event days by arm under common-horizon truncation and under capping at the patient-specific last-contact day (LASTCTDY), on the 520-person cohort. The observed-to-horizon rows restrict to patients whose last contact reaches the window ($n = 213$ chemotherapy and 219 panitumumab at day 84; 194 and 203 at day 120). Capping applies only to intervals with complete boundaries, so these are diagnostics bounding, not removing, the unresolved-record problem discussed above. The day-84 and day-120 rows apply the same last-contact cap, which is why they sit slightly below the plain uncapped truncation differences quoted in the text (3.08 against 3.14 at day 84 and 4.03 against 4.10 at day 120).

Window	Chemotherapy	Panitumumab	Difference
Last contact capped	19.18	30.90	11.73
Day 84, all patients	11.25	14.33	3.08
Day 84, observed to day 84	11.25	14.93	3.68
Day 120, all patients	15.55	19.58	4.03
Day 120, observed to day 120	16.12	21.25	5.13

on why patients discontinued, and the supplied extract does not provide a treatment-stop reason that resolves this. The per-exposure quantities are therefore reported as descriptive rates per treatment-day, and the truncation table as a diagnostic showing that the effect is exposure-dependent. The v2 disposition file makes a formal fixed-horizon analysis possible in principle, but it does not remove the need to define censoring, competing events, and the treatment policy. The present data support a last-contact-capped observed-window diagnostic; a fixed-horizon restricted mean remains a follow-up analysis rather than a replacement for the randomized estimand.

5.4 Heterogeneity and Specification Sensitivity

Subgroup effects are formed draw by draw, averaging unit-level effects within each posterior draw over the members of a subgroup, so that a contrast between two subgroups inherits their posterior covariance rather than being read off two unrelated marginal intervals. Figure 1 displays the result.

Posterior means are positive in every prespecified subgroup on both endpoints, but no ZIC-BCF-Smear pairwise contrast has a 95% credible interval excluding zero. The largest ZIC-BCF-Smear signal, ECOG 0 versus ECOG 1, is 6.28 event-days $[-2.57, 22.20]$ on the summed endpoint and 6.69 days $[-0.83, 20.43]$ on the distinct-day endpoint. The Gamma hurdle benchmark reaches the opposite conclusion on the same data, estimating those contrasts at 26.94 event-days $[4.74, 51.94]$ and 23.34 days $[7.83, 40.98]$, reproducibly across seeds, and it likewise returns credible participation-margin contrasts for ECOG 0 versus ECOG 1 and for HPV-negative versus HPV-not-recorded where ZIC-BCF-Smear returns none. Because both models are fit to identical data, covariates, and treatment coding, the difference is attributable to specification: the parametric Gamma positive component with linear dummy-encoded covariates extrapolates subgroup means without the regularization the causal forest imposes. This is the applied

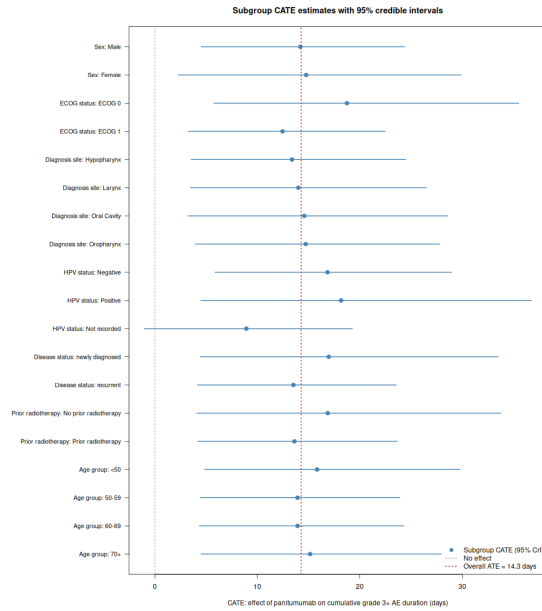


Figure 1: ZIC-BCF-Smear subgroup conditional average treatment effects on summed severe-adverse-event-record duration, in event-days, with 95% credible intervals; the dashed line marks no effect and the dotted line the overall effect of 14.3 event-days. Every subgroup posterior mean falls within roughly five event-days of the overall effect and every interval covers it, so the estimated effect surface is broadly homogeneous. Interval width tracks subgroup size almost monotonically, widest for HPV-positive patients ($n = 69$), women ($n = 65$), and patients aged 70 or older ($n = 36$), which is the expected consequence of borrowing strength across a tree ensemble rather than fitting stratum-specific models.

counterpart of the divergence documented under link-nonlinearity in Section 4.4, where the same benchmark, fit to data whose truth was known, produced negative CATE correlations and diverged outright on 4 to 5 seeds in 100. Its practical consequence here is concrete, since an ECOG effect-modification claim would be reported as credible under the parametric benchmark and as inconclusive under ZIC-BCF-Smear.

Two of these contrasts deserve to be named for what they are. The HPV comparison that the benchmark reports as credible is against a category created by coalescing patients with no recorded baseline HPV result, so it is a missingness contrast over 181 patients rather than a comparison of two measured biological states, and it should not be read as evidence about HPV biology. Prior radiotherapy is the clinically motivated candidate modifier: the summed-duration effect is 16.87 event-days [4.10, 33.76] among the 104 patients without prior radiotherapy and 13.62 [4.17, 23.65] among the 416 with prior radiotherapy, with contrast 3.25 [−7.61, 17.82]. This contrast is not credible, and the smaller stratum is limited. The composite prior-treatment variable gives materially identical strata (the two indicators differ for four patients), so it is not reported as a separate subgroup comparison, which would add a near-duplicate contrast to an analysis that is already unadjusted for multiplicity. The radiotx domain contains records for all 520 subjects, each patient recorded as PRRADIO=N carrying an explicit RAANY=N record, so its subject-level agreement with PRRADIO is a genuine coding concordance rather than an artifact of absent files being coalesced to no radiotherapy. The lesion-level flag LSPIRRNY is not merely incomplete relative to PRRADIO but frequently contradicts it: 42 of the 104 patients recorded as having had no prior radiotherapy (40.4%) have at least one lesion flagged as previously irradiated, against 110 of the 416 patients recorded as irradiated (26.4%), and among the 373 patients with any lesion-level information the corresponding rates are 50.0% and 38.1%. A flag that is more often positive among patients recorded as not irradiated is either a different construct, such as a lesion treated in another field or for another indication, or a merge or level-coding problem; it is therefore not used as a covariate and is reported here rather than passed over in silence.

Which answer is right cannot be settled internally, but it can be settled against the literature. Previous SPECTRUM subgroup work concentrated on efficacy and on HPV or p16 status, and the one available HPV-specific safety analysis is a conference abstract that does not estimate baseline-modified causal effects on either burden outcome (Stöhlmacher-Williams et al., 2012); neither ECOG nor HPV status is established anywhere in the SPECTRUM literature as a modifier of the causal burden of panitumumab-associated toxicity (Vermorken et al., 2013; Stöhlmacher-Williams et al., 2012). A regularized estimator returning no credible heterogeneity is therefore the concordant answer, and the Gamma-specific signals are best read as model-sensitive hypotheses rather than as externally corroborated effect modification. This is the same asymmetry the simulations produced, with the direction of the error preserved: under functional misspecification the parametric hurdle does not merely lose precision, it reports structure that is not there.

5.5 A Second Application: Dental Expenditures

The oncology study exercises the estimator where the applied literature is thin, since the study is novel in exploring effect in adverse-event duration scale; consequently, external agreement can be assessed for direction and robustness but not for magnitude. A second study supplies the complementary case, estimating the causal effect of dental-insurance coverage on annual dental expenditure in the 2023 Medical Expenditure Panel Survey ([Agency for Healthcare Research and Quality, 2023](#)), an $N = 18,763$ observational sample in which 54.9% of respondents report zero dental spending and the positive part is severely right-skewed (median \$361, mean \$1,038, maximum \$40,350). This is the canonical semicontinuous outcome of health economics ([Duan et al., 1983](#); [Manning and Mullahy, 2001](#)), and decades of applied work have documented the gradients that coverage effects should exhibit by race and ethnicity, age, income, and region ([Manski et al., 2002](#); [Zhang, 2016](#); [Reda et al., 2018](#); [Vujicic et al., 2016](#)). It is reported in full in Section 6 of the Supplementary Material ([Souto et al., 2026](#)) rather than here, because its role is confirmatory rather than methodological.

MEPS is a stratified, clustered, unequal-probability sample, so two estimands are reported for every quantity and they answer different questions. The analytic-sample average describes the 18,763 records, which oversample several subpopulations; the survey-weighted population target uses the person weight and carries a design-based variance component obtained by Taylor linearization over the sampling strata and primary sampling units ([Binder, 1983](#); [Agency for Healthcare Research and Quality, 2025](#)), and only that column supports a reading about the United States population. Clustering ordinarily widens an interval and stratification ordinarily narrows it, so which dominates is a matter for computation rather than assertion. Race and ethnicity are taken from the standard MEPS analytic variable, whose first level is Hispanic; the alternative race-only variable carries no ethnicity dimension and, in this sample, would place 3,758 Hispanic respondents inside the White category.

Its value as a check is that ZIC-BCF-Smear recovers the documented gradients directly on the dollar-scale conditional effect, and the parametric benchmark, fit to the same sample with the same covariates, does not. The estimated average effect of coverage is \$188.02 [\$156.95, \$220.67] as a sample average and \$184.67 [\$154.63, \$214.70] as a design-aware population target, positive as the MEPS literature on coverage, visits, and expenditure would require ([Manski et al., 2002](#); [Christian and Chattopadhyay, 2014](#); [Naib et al., 2026](#)). The design component is small here, contributing a standard error of \$1.36 against a posterior standard deviation of \$15.26, so the interval widens only slightly; that is a result rather than an assumption, and it did not have to come out that way.

On the conditional effects, the contrast between non-Hispanic White and non-Hispanic Black beneficiaries is +\$71.42 [\$11.96, \$130.87] and the contrast between Hispanic and non-Hispanic White beneficiaries is −\$62.38 [−\$100.02, −\$24.74], reproducing the racial and ethnic gradient documented across the oral-health disparities literature ([Zhang, 2016](#); [Reda et al., 2018](#); [Robison et al., 2020](#)); the effect rises monotonically across the five age groups from \$127 to \$250, with the children-versus-65-and-over contrast at −\$123.30 [−\$204.88, −\$41.73], tracking the documented age gradient in dental expenditure and

coverage gaps (Christian and Chattopadhyay, 2014; Zhang, 2016; Manski et al., 2015); and the low-income-versus-middle-income contrast is $-\$79.08$ [$-\$145.02, -\13.14], matching the finding that coverage is a weaker enabling resource where cost-sharing and annual caps bite hardest (Manski et al., 2002; Reda et al., 2018; Li and Yao, 2021; Vujicic et al., 2016). In total 18 of the 56 pairwise contrasts are credible under design-aware inference.

Two results changed direction under the corrections and are reported as such. The South-versus-West regional contrast, previously credible, is $-\$41.12$ [$-\$87.62, \5.38] once the survey design is respected, and no longer excludes zero; it was the least literature-anchored of the reported gradients (Luo et al., 2021; Joudi et al., 2024) and should now be described as suggestive only. And the average effect is no longer roughly half the unadjusted insured-versus-uninsured spending gap, because that gap is $\$185.9$, essentially the adjusted estimate itself. The attenuation the selection story predicts appears instead on the participation margin, where the adjusted effect of 0.139 [0.124, 0.155] is about seven tenths of the unadjusted 0.201 difference in the probability of any dental spending. On the dollar scale the observed covariates absorb almost none of the raw association, which is more naturally read as evidence that they fail to capture the latent oral-health demand driving selection into coverage than as evidence that no such selection exists. That is a limitation of the adjustment set rather than a strength of the estimate, and Section 6 of the Supplementary Material treats it as one.

The comparison with the benchmark is not symmetric. Under design-aware inference the Gamma hurdle produces exactly one credible dollar-scale contrast out of 56, and it runs against the literature rather than with it, placing the low-income tertile $\$124.06$ [$\$2.80, \245.43] *above* the high-income tertile. Its race and ethnicity point estimates do not order the groups as three decades of applied work do (non-Hispanic Black $\$144.2$ above non-Hispanic White $\$143.3$ above Hispanic $\$138.4$) (Zhang, 2016; Reda et al., 2018; Robison et al., 2020), its dollar-scale age gradient is non-monotone, and its regional ordering is inverted, with the South highest at $\$198.7$. On the estimand that both models are asked to report, the conditional effect on the dollar scale, ZIC-BCF-Smear converges to the established applied literature and the Gamma hurdle does not. The mechanism is the one identified in Section 4.4: the Gamma positive-intensity likelihood imposes a single parametric tail shape that smooths away the heavy right tail in which the literature-aligned heterogeneity lives, whereas the nonparametric smearing re-transformation absorbs it. What rules out the reading that the Gamma hurdle is simply measuring something different is that it does recover credible structure on its participation margin, where its decomposition is well suited, returning 17 credible contrasts to ZIC-BCF-Smear’s 11. The parametric benchmark loses the heterogeneity exactly where its distributional assumption binds, which is the scale-misspecification failure of Section 3.5 observed against an external standard rather than a known truth. Taken with the simulations, the applied evidence points one way: on hurdle-structured data the proposed estimator recovers heterogeneity that the classical parametric hurdle either misses or inverts.

6 Limitations

The mechanisms behind the empirical performance of ZIC-BCF-Smear have been developed alongside the results they explain, in Sections 4.3, 4.4, and 4.6 for the simulations and in Section 5 for the two applications. This section is confined to what the framework does not resolve.

While ZIC-BCF-Smear demonstrates strong empirical performance, it is subject to several limitations that warrant candid acknowledgment.

First, a *mild under-coverage of CATE credible intervals* is observed under complex, non-linear conditional mean surfaces at larger sample sizes (specifically, 89.7% and 91.7% coverage for $N = 500$, and similar behavior at $N = 1000$). This coverage dip occurs because the BCF prognostic and treatment forests incorporate highly conservative regularizing priors designed to prevent overfitting. Under highly complex, non-linear link surfaces, these priors can cause the posterior distributions to be overly conservative (shrunk toward the prior mean), leading to slightly narrower credible intervals that do not fully cover the true CATE for units in the tails of the covariate distribution.

Second, the continuous intensity stage (Stage 2) assumes *homoskedastic log-scale errors* ($\epsilon_i \sim \mathcal{N}(0, \sigma_c^2)$), and Duan’s smearing factor inherits that assumption: a single scalar per posterior draw is applied to both counterfactual arms, so every response-scale estimate scales linearly in it. Because the assumption is consequential it is now tested rather than asserted. The `zicbcf` package reports a studentized Breusch–Pagan test of the log-scale residuals against the fitted mean and the covariates (Breusch and Pagan, 1979; Koenker, 1981), computed both at the posterior mean and within every retained draw, together with a locally smeared sensitivity analysis that replaces the scalar with stratum-specific factors over quantiles of the fitted log-scale mean. The two applications bracket the possibilities. In the SPECTRUM cohort the test gives $p = 0.12$ on the summed endpoint and $p = 0.16$ on the distinct-day endpoint, while locally smeared effects remain close to the scalar-smearing results. In MEPS the test rejects about as decisively as a test can, at $p = 1.2 \times 10^{-24}$ and in every one of the 200 draws examined, with the residual standard deviation rising monotonically from 1.01 to 1.20 across strata of the fitted mean; yet replacing the scalar with stratum-specific factors moves the average effect only from \$187 to \$193, a change of 3% against a posterior standard deviation of \$16. The simulation grid reproduces both regimes and adds the case that matters most, since under the Tweedie generators the test rejects in every replicate and the locally smeared estimate moves by 18% to 25%. The practical guidance that follows is that the test should be run and the locally smeared sensitivity reported, because the assumption can fail without the estimate moving, and a rejected test is therefore not by itself a reason to discard the fit.

Third, the Subpopulation Propensity Adjustment (SPA) framework conditions on the post-treatment event $I_i = 1$ ($Y_i > 0$). This causal identification strategy relies on the *subpopulation strong ignorability assumption* (Assumption 3.1). Although this assumption is mathematically guaranteed under full-sample strong ignorability and conditional two-part independence, it may be violated in settings where unobserved confounders simultaneously affect both the hurdle participation process and the continuous response

intensity. Under selection on unobservables, the active-subset regression may suffer from residual collider bias that cannot be adjusted for via the subpopulation propensity score alone.

Fourth, both propensity scores are *plug-in estimates*. As described in Section 3.6, $\hat{\pi}_i$ and $\hat{\pi}_i^+$ are fit once by logistic regression before the sampler starts and are then held fixed, so their estimation uncertainty is cut out of the outcome posterior. This is the same design choice made by BCF itself (Hahn et al., 2020) and is deliberate, since feeding the propensity score into a joint posterior would allow the flexible outcome model to distort the treatment model. The cost is a mild understatement of posterior uncertainty in settings with weak overlap, where the propensity model is itself poorly determined, and a sensitivity to misspecification of the propensity model that the credible intervals do not display.

Fifth, the estimator targets a contrast in the outcome as recorded and has *no mechanism for standardizing exposure or follow-up time*. When the outcome accrues over time and the intervention alters how long patients remain on treatment, as in the SPECTRUM application, treatment duration lies on the causal pathway and the estimand is the effect of assignment under the trial’s own differential duration. That is a coherent target, and under randomization it is identified, but it is not a fixed-horizon restricted mean and it is not a rate per unit of exposure. Section 5.3 shows how far the two can diverge in practice. The v2 disposition domains provide patient-level safety-follow-up and last-contact days, but only 354 patients completed the safety follow-up and last contact is not equivalent to complete safety ascertainment. Extending the two-part construction to a fixed-horizon or multistate formulation therefore requires an explicit censoring rule, treatment policy, and competing-event definition, and remains the most consequential extension for clinical safety endpoints.

Sixth, the sampler has *no interface for survey weights, strata, or primary sampling units*. In the MEPS application the design is therefore handled outside the model: unit-level effect draws are averaged with the person weight to obtain population targets, and a design-based variance component from the sampling strata and primary sampling units is added to the posterior variance (Binder, 1983). This is defensible because every reported quantity is a linear functional of unit-level effects, but it treats sampling and posterior uncertainty as approximately independent and does not let the design influence the fitted response surface itself. A weighted or design-informed prior would be the principled alternative and is not attempted here.

Finally, due to the *extreme computational complexity of the enriched Dirichlet process benchmark (EDP+ZI)*, its inclusion in the main Monte Carlo simulation grid was not possible. EDP+ZI requires $O(N^3)$ matrix operations at each Gibbs sweep to update the Dirichlet process parameters, leading to a $\approx 90\times$ computational slowdown compared to ZIC-BCF-Smear (≈ 27.5 hours versus ≈ 18 minutes for a single standard fit at $N = 500$), as documented in Section 4.1. While this computational bottleneck is itself a practical argument for the forest-based approach, the absence of a direct empirical performance comparison against EDP+ZI across the sensitivity cells remains a limitation.

7 Conclusion

This paper introduced the Zero-Inflated Continuous Bayesian Causal Forest with Duan’s Smearing (ZIC-BCF-Smear), a two-part nonparametric framework for estimating conditional and average treatment effects on semicontinuous outcomes. By decoupling the causal process into a Probit BCF hurdle stage and a Gaussian BCF intensity stage fit strictly on active units, it avoids the computational burden and data augmentation noise that degrade joint selection models; Subpopulation Propensity Adjustment addresses the selection bias inherent in conditioning on active participation; and a nonparametric smearing estimator maps predictions back to the response scale, bypassing the fragile parametric multipliers that cause scaling bias and numerical instability in traditional two-part models.

Comprehensive simulation studies show that ZIC-BCF-Smear matches or outperforms correctly specified parametric hurdle models without committing to any parametric family, and remains stable and accurate under nonlinear surfaces where those benchmarks diverge numerically and reverse the ordering of treatment effects. Two applied studies carry it onto real data. In the SPECTRUM randomized phase III head-and-neck cancer trial, assignment to panitumumab plus chemotherapy credibly increases severe-toxicity burden by about ten days per patient once concurrent adverse events are counted once, and the two-part decomposition places that effect mainly in the positive-intensity part rather than in whether the reconstructed severe-AE burden is positive. That effect is the effect of assignment under the trial’s own differential treatment duration, which the study reports explicitly rather than leaving implicit. In the 2023 Medical Expenditure Panel Survey, where a mature applied literature supplies an external standard, the estimator recovers the documented dental-coverage gradients by race and ethnicity, age, and income directly on the expenditure scale, and does so under survey-weighted, design-aware inference, while a parametric Gamma hurdle benchmark fit to the same data reproduces them only on its participation margin.

Two extensions follow directly from the limitations. Heteroskedastic log-scale errors could be accommodated by modeling the residual variance nonparametrically as a separate regression forest, and the framework could be generalized to longitudinal or survival settings with right-censoring. The oncology application motivates the latter most directly, since a prespecified fixed-horizon restricted-mean or multistate formulation would sharpen the safety estimand by standardizing follow-up time and accommodating competing events.

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This publication is based on research using information obtained from www.projectdatasphere.org, which is maintained by Project Data Sphere. Neither Project Data Sphere nor the owner(s) of any information from the web site have contributed to, approved or are in any way responsible for the contents of this publication.

Data Availability

The health-economics application uses the 2023 Medical Expenditure Panel Survey Full-Year Consolidated file (HC-251), a public release of the Agency for Healthcare Research and Quality ([Agency for Healthcare Research and Quality, 2023](#)). It is freely redistributable subject to two conditions of use that survive redistribution, namely that the data be used for statistical reporting and analysis only and that they not be linked with individually identifiable records from any source other than the Medical Expenditure Panel Survey and the National Health Interview Survey from which its sample is drawn. That arm of the study is fully reproducible: the analysis repository accompanying the `zicbcf` package ([Souto, 2026](#)) contains a download-and-prepare script for the public file together with every analysis script and derived results table.

The oncology application uses individual-participant data from the SPECTRUM trial obtained through Project Data Sphere ([Project Data Sphere, 2026](#)). Those data cannot be redistributed. Section III.B(ii) of the Project Data Sphere Platform Agreement prohibits disclosure to any third party who is not a registered Authorized User, and Article VIII requires that the data be held in strict confidence ([Project Data Sphere, 2020](#)). This is a contractual confidentiality obligation rather than a patient-privacy one, so further de-identification does not make a per-patient file redistributable. Article VII permits disclosure of what is reasonably necessary to publish research results, which is the basis on which the aggregate estimates, tables, and figures reported here appear; per-patient derivatives, including patient-level endpoint files and unit-level posterior draws, are withheld. Access is granted per individual, so any reader wishing to reproduce this arm must register as an Authorized User and request the `HeadNe_Amgen_2007_265` package directly; the scripts also document the v2 demo, radiotherapy, lesion, and disposition domains used for the updated diagnostics.

Supplementary Material

Supplement to “Distribution-Free Causal Inference for Semicontinuous Outcomes: A Two-Part Bayesian Causal Forest with Duan’s Smearing”. A single PDF in six sections: 1. the six data-generating processes with closed-form true potential outcomes; 2. the proof of Lemma 3.3, the posterior sampling scheme, the fitting algorithm, tree priors, the MCMC convergence diagnostics, and the test of the homoskedasticity assumption behind the smearing re-transformation; 3. the timing analysis behind the exclusion of the enriched Dirichlet process benchmark; 4. the complete simulation results, sensitivity figures, and numerical-stability and hurdle-stage tables; 5. the extended oncology applied study on the SPECTRUM trial, including the treatment-exposure and outcome-completeness diagnostics and the data-sharing position; and 6. the extended applied study on dental expenditure in the 2023 Medical Expenditure Panel Survey.

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